MAT627 Chapter 3 Programming Assignment

Emre Tekmen

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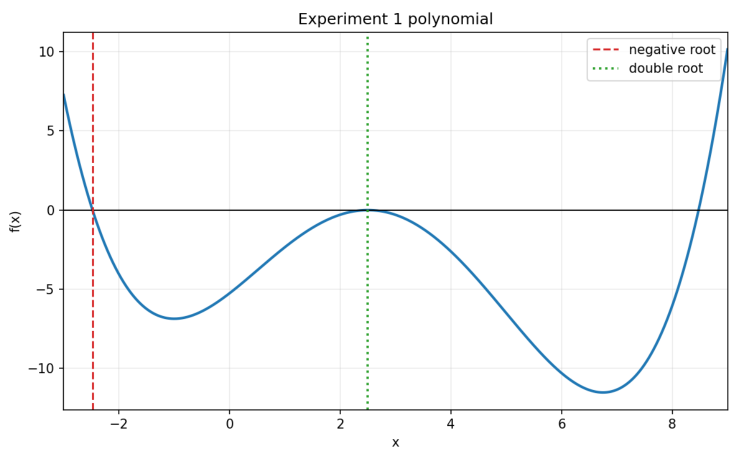

Figure 1: Experiment 1 polynomial on $[-3, 9]$. The negative root and the double root are marked from the computed iterates.

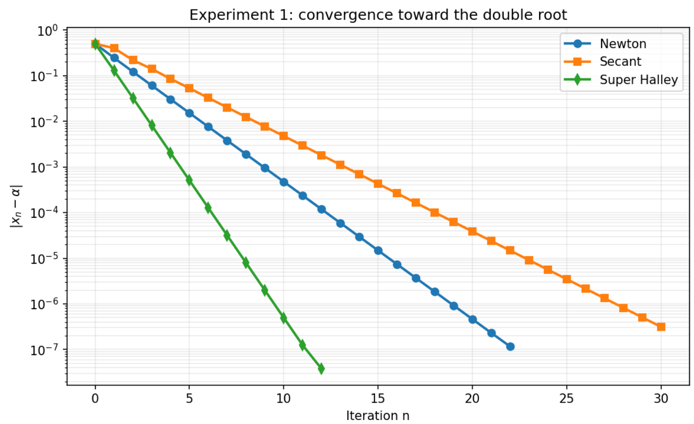

Figure 2: Experiment 1: convergence histories for Newton, secant, and Super Halley near the double root $\alpha = 2.5$.

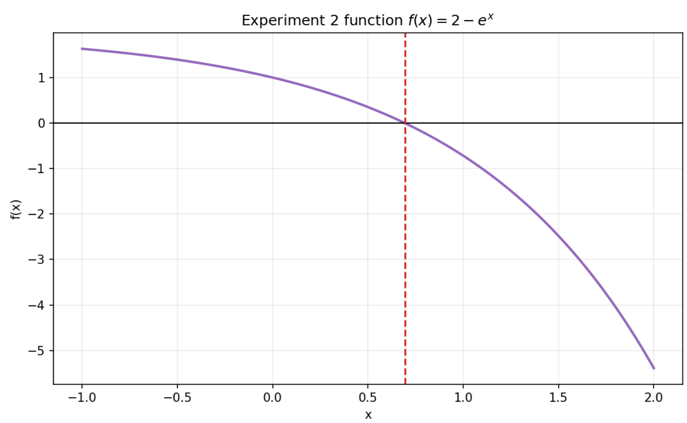

Figure 3: Experiment 2 function $f(x) = 2 - e^x$ on $[-1, 2]$.

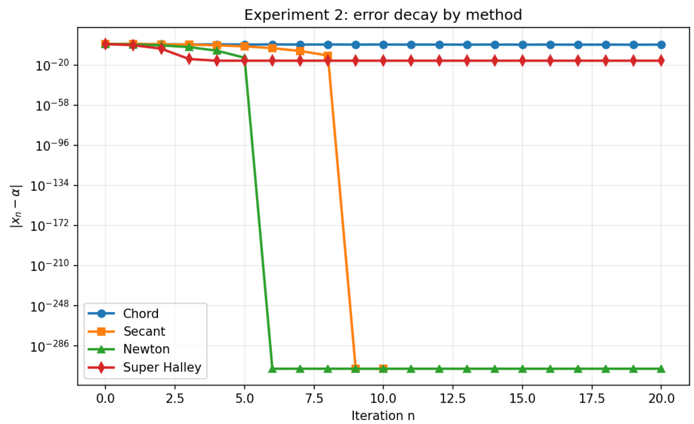

Figure 4: Experiment 2: error decay for the chord, secant, Newton, and Super Halley methods.

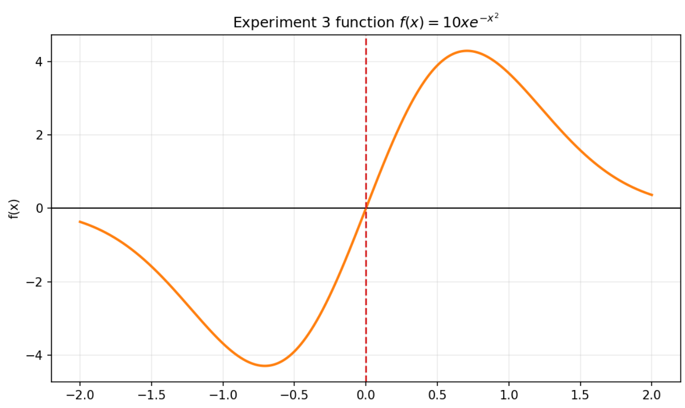

Figure 5: Experiment 3 function $f(x) = 10xe^{-x^2}$ on $[-2, 2]$.

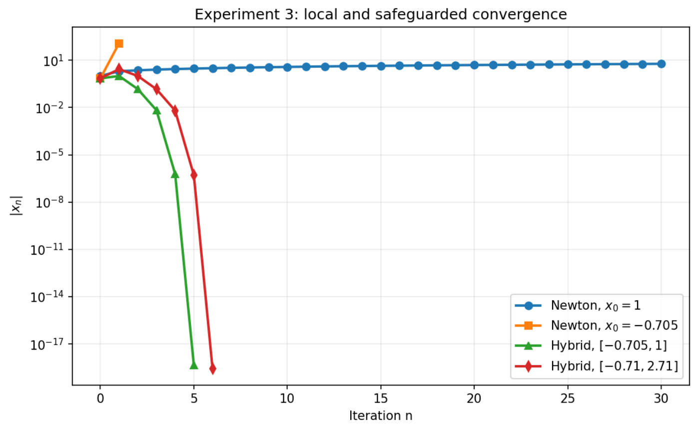

Figure 6: Experiment 3: Newton's sensitivity near a point where $f'(x) \approx 0$ versus the safeguarded hybrid iterations.

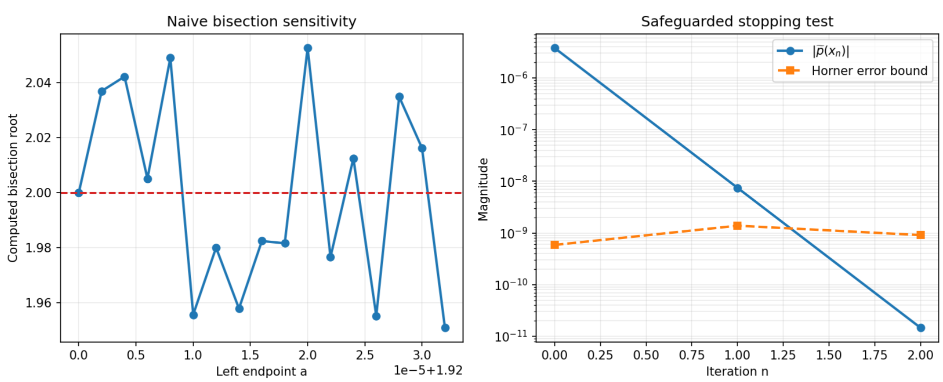

Figure 7: Graduate section: interval sensitivity of naive bisection for $(x-2)^9$ and the Horner error-bound stopping test.

1 Experiment 1

Unknown Negative Root

Bisection

Table 1: Bisection on $[-3, -2]$ for the polynomial in Experiment 1.

n	a_n	b_n	x_n	$\frac{1}{2}(b_n - a_n)$
0	-3.000000×10^0	-2.000000×10^0	-2.500000×10^0	5.000000×10^{-1}
1	-2.500000×10^0	-2.000000×10^0	-2.250000×10^0	2.500000×10^{-1}
2	-2.500000×10^0	-2.250000×10^0	-2.375000×10^0	1.250000×10^{-1}
3	-2.500000×10^0	-2.375000×10^0	-2.437500×10^0	6.250000×10^{-2}
4	-2.500000×10^0	-2.437500×10^0	-2.468750×10^0	3.125000×10^{-2}
5	-2.500000×10^0	-2.468750×10^0	-2.484375×10^0	1.562500×10^{-2}
6	-2.484375×10^0	-2.468750×10^0	-2.476562×10^0	7.812500×10^{-3}
7	-2.484375×10^0	-2.476562×10^0	-2.480469×10^0	3.906250×10^{-3}
8	-2.480469×10^0	-2.476562×10^0	-2.478516×10^0	1.953125×10^{-3}
9	-2.478516×10^0	-2.476562×10^0	-2.477539×10^0	9.765625×10^{-4}
10	-2.477539×10^0	-2.476562×10^0	-2.477051×10^0	4.882812×10^{-4}
11	-2.477539×10^0	-2.477051×10^0	-2.477295×10^0	2.441406×10^{-4}
12	-2.477295×10^0	-2.477051×10^0	-2.477173×10^0	1.220703×10^{-4}
13	-2.477295×10^0	-2.477173×10^0	-2.477234×10^0	6.103516×10^{-5}
14	-2.477234×10^0	-2.477173×10^0	-2.477203×10^0	3.051758×10^{-5}
15	-2.477234×10^0	-2.477203×10^0	-2.477219×10^0	1.525879×10^{-5}
16	-2.477234×10^0	-2.477219×10^0	-2.477226×10^0	7.629395×10^{-6}
17	-2.477226×10^0	-2.477219×10^0	-2.477222×10^0	3.814697×10^{-6}
18	-2.477226×10^0	-2.477222×10^0	-2.477224×10^0	1.907349×10^{-6}
19	-2.477226×10^0	-2.477224×10^0	-2.477225×10^0	9.536743×10^{-7}

Status: `converged`

Newton

Table 2: Newton's method from $x_0 = -2$ for the negative root.

n	$\lvert x_n - x_{n-1}\rvert$	$\lvert f(x_n)\rvert$	x_n	Rate
0	–	4.050000×10^0	-2.000000×10^0	–
1	6.428571×10^{-1}	1.948588×10^0	-2.642857×10^0	–
2	1.534610×10^{-1}	1.329043×10^{-1}	-2.489396×10^0	–
3	1.209810×10^{-2}	7.864386×10^{-4}	-2.477298×10^0	1.773432
4	7.244502×10^{-5}	2.809329×10^{-8}	-2.477226×10^0	2.014635
5	2.588077×10^{-9}	8.881784×10^{-16}	-2.477226×10^0	2.000726

Status: `converged`

Secant

Table 3: Secant method from $x_0 = -3$, $x_1 = -2$ for the negative root.

n	$\lvert x_n - x_{n-1}\rvert$	$\lvert f(x_n)\rvert$	x_n	Rate
0	–	7.260000×10^{0}	-3.000000×10^{0}	–
1	1.000000×10^{0}	4.050000×10^{0}	-2.000000×10^{0}	–
2	3.580902×10^{-1}	1.218635×10^{0}	-2.358090×10^{0}	–
3	1.541239×10^{-1}	3.863850×10^{-1}	-2.512214×10^{0}	0.820888
4	3.710307×10^{-2}	2.292941×10^{-2}	-2.475111×10^{0}	1.689217
5	2.078480×10^{-3}	3.916705×10^{-4}	-2.477189×10^{0}	2.023840
6	3.612071×10^{-5}	4.087090×10^{-7}	-2.477226×10^{0}	1.406120
7	3.765274×10^{-8}	7.274181×10^{-12}	-2.477226×10^{0}	1.694305

Status: `converged`

Hybrid

Table 4: Hybrid method on $[-3, -2]$ for the negative root.

n	a_n	b_n	x_n	$\frac{1}{2}(b_n - a_n)$
0	-3.000000×10^{0}	-2.000000×10^{0}	-3.000000×10^{0}	5.000000×10^{-1}
1	-3.000000×10^{0}	-2.000000×10^{0}	-2.000000×10^{0}	5.000000×10^{-1}
2	-2.642857×10^{0}	-2.000000×10^{0}	-2.642857×10^{0}	3.214286×10^{-1}
3	-2.489396×10^{0}	-2.000000×10^{0}	-2.489396×10^{0}	2.446981×10^{-1}
4	-2.477298×10^{0}	-2.000000×10^{0}	-2.477298×10^{0}	2.386490×10^{-1}
5	-2.477226×10^{0}	-2.000000×10^{0}	-2.477226×10^{0}	2.386128×10^{-1}
6	-2.477226×10^{0}	-2.000000×10^{0}	-2.477226×10^{0}	2.386128×10^{-1}

Status: `converged`

Super Halley

Table 5: Super Halley method from $x_0 = -2$ for the negative root.

n	$\lvert x_n - x_{n-1}\rvert$	$\lvert f(x_n)\rvert$	x_n	Rate
0	–	4.050000×10^{0}	-2.000000×10^{0}	–
1	4.943155×10^{-1}	1.870768×10^{-1}	-2.494316×10^{0}	–
2	1.709031×10^{-2}	4.075124×10^{-6}	-2.477225×10^{0}	–
3	3.754184×10^{-7}	8.881784×10^{-16}	-2.477226×10^{0}	3.187833
4	0.000000×10^{0}	8.881784×10^{-16}	-2.477226×10^{0}	–

Status: `converged`

Double Root at $\alpha = 2.5$

Newton

Table 6: Newton's method from $x_0 = 2$ for the double root.

n	$\|x_n - x_{n-1}\|$	$\|f(x_n)\|$	x_n	$\|x_n - \alpha\|$	Rate
0	–	2.900000×10^{-1}	2.000000×10^{0}	5.000000×10^{-1}	–
1	2.543860×10^{-1}	7.104999×10^{-2}	2.254386×10^{0}	2.456140×10^{-1}	–
2	1.235756×10^{-1}	1.764154×10^{-2}	2.377962×10^{0}	1.220384×10^{-1}	0.983933
3	6.117603×10^{-2}	4.398467×10^{-3}	2.439138×10^{0}	6.086238×10^{-2}	0.994703
4	3.046622×10^{-2}	1.098315×10^{-3}	2.469604×10^{0}	3.039616×10^{-2}	0.997957
5	1.520633×10^{-2}	2.744275×10^{-4}	2.484810×10^{0}	1.518983×10^{-2}	0.999123
6	7.596914×10^{-3}	6.858867×10^{-5}	2.492407×10^{0}	7.592916×10^{-3}	0.999597
7	3.796950×10^{-3}	1.714494×10^{-5}	2.496204×10^{0}	3.795966×10^{-3}	0.999807
8	1.898105×10^{-3}	4.285958×10^{-6}	2.498102×10^{0}	1.897861×10^{-3}	0.999906
9	9.489608×10^{-4}	1.071455×10^{-6}	2.499051×10^{0}	9.489000×10^{-4}	0.999953
10	4.744576×10^{-4}	2.678595×10^{-7}	2.499526×10^{0}	4.744424×10^{-4}	0.999977
11	2.372231×10^{-4}	6.696435×10^{-8}	2.499763×10^{0}	2.372193×10^{-4}	0.999988
12	1.186101×10^{-4}	1.674102×10^{-8}	2.499881×10^{0}	1.186092×10^{-4}	0.999994
13	5.930471×10^{-5}	4.185246×10^{-9}	2.499941×10^{0}	5.930448×10^{-5}	0.999997
14	2.965226×10^{-5}	1.046311×10^{-9}	2.499970×10^{0}	2.965221×10^{-5}	0.999998
15	1.482612×10^{-5}	2.615774×10^{-10}	2.499985×10^{0}	1.482609×10^{-5}	1.000000
16	7.413051×10^{-6}	6.539391×10^{-11}	2.499993×10^{0}	7.413040×10^{-6}	0.999999
17	3.706501×10^{-6}	1.634870×10^{-11}	2.499996×10^{0}	3.706539×10^{-6}	0.999992
18	1.853266×10^{-6}	4.086509×10^{-12}	2.499998×10^{0}	1.853273×10^{-6}	1.000005
19	9.264803×10^{-7}	1.022293×10^{-12}	2.499999×10^{0}	9.267926×10^{-7}	0.999760
20	4.634641×10^{-7}	2.557954×10^{-13}	2.500000×10^{0}	4.633285×10^{-7}	1.000454
21	2.319673×10^{-7}	6.306067×10^{-14}	2.500000×10^{0}	2.313612×10^{-7}	1.001677
22	1.145226×10^{-7}	1.598721×10^{-14}	2.500000×10^{0}	1.168386×10^{-7}	0.983772

Status: `converged; warning:  |f'(x_n)| < TOL`

Secant

Status: `converged`

Super Halley

Status: `converged; warning:  |f'(x_n)| < TOL`

2 Experiment 2

Chord

Status: `fixed-iteration run complete`

Table 7: Secant method from $x_0 = 2$, $x_1 = 2.1$ for the double root.

n	$\lvert x_n - x_{n-1} \rvert$	$\lvert f(x_n) \rvert$	x_n	$\lvert x_n - \alpha \rvert$	Rate
0	–	2.900000×10^{-1}	2.000000×10^{0}	5.000000×10^{-1}	–
1	1.000000×10^{-1}	1.868160×10^{-1}	2.100000×10^{0}	4.000000×10^{-1}	–
2	1.810513×10^{-1}	5.653507×10^{-2}	2.281051×10^{0}	2.189487×10^{-1}	2.700626
3	7.856676×10^{-2}	2.332523×10^{-2}	2.359618×10^{0}	1.403819×10^{-1}	0.737555
4	5.518207×10^{-2}	8.611379×10^{-3}	2.414800×10^{0}	8.519984×10^{-2}	1.123510
5	3.229567×10^{-2}	3.324397×10^{-3}	2.447096×10^{0}	5.290417×10^{-2}	0.954243
6	2.030717×10^{-2}	1.263021×10^{-3}	2.467403×10^{0}	3.259701×10^{-2}	1.016252
7	1.244236×10^{-2}	4.830553×10^{-4}	2.479845×10^{0}	2.015464×10^{-2}	0.992822
8	7.705911×10^{-3}	1.843372×10^{-4}	2.487551×10^{0}	1.244873×10^{-2}	1.002143
9	4.755274×10^{-3}	7.041687×10^{-5}	2.492307×10^{0}	7.693456×10^{-3}	0.998822
10	2.939348×10^{-3}	2.689152×10^{-5}	2.495246×10^{0}	4.754108×10^{-3}	1.000234
11	1.816034×10^{-3}	1.027139×10^{-5}	2.497062×10^{0}	2.938074×10^{-3}	0.999780
12	1.122326×10^{-3}	3.923118×10^{-6}	2.498184×10^{0}	1.815748×10^{-3}	1.000004
13	6.935771×10^{-4}	1.498471×10^{-6}	2.498878×10^{0}	1.122171×10^{-3}	0.999949
14	4.286418×10^{-4}	5.723555×10^{-7}	2.499306×10^{0}	6.935288×10^{-4}	0.999989
15	2.649081×10^{-4}	2.186186×10^{-7}	2.499571×10^{0}	4.286207×10^{-4}	0.999986
16	1.637200×10^{-4}	8.350438×10^{-8}	2.499735×10^{0}	2.649007×10^{-4}	0.999994
17	1.011836×10^{-4}	3.189573×10^{-8}	2.499836×10^{0}	1.637171×10^{-4}	0.999995
18	6.253457×10^{-5}	1.218306×10^{-8}	2.499899×10^{0}	1.011825×10^{-4}	0.999997
19	3.864836×10^{-5}	4.653507×10^{-9}	2.499937×10^{0}	6.253414×10^{-5}	0.999998
20	2.388594×10^{-5}	1.777480×10^{-9}	2.499961×10^{0}	3.864820×10^{-5}	0.999998
21	1.476231×10^{-5}	6.789378×10^{-10}	2.499976×10^{0}	2.388590×10^{-5}	0.999999
22	9.123623×10^{-6}	2.593294×10^{-10}	2.499985×10^{0}	1.476227×10^{-5}	1.000003
23	5.638649×10^{-6}	9.905499×10^{-11}	2.499991×10^{0}	9.123625×10^{-6}	0.999988
24	3.484874×10^{-6}	3.783640×10^{-11}	2.499994×10^{0}	5.638751×10^{-6}	0.999994
25	2.153841×10^{-6}	1.445155×10^{-11}	2.499997×10^{0}	3.484910×10^{-6}	1.000033
26	1.331047×10^{-6}	5.520029×10^{-12}	2.499998×10^{0}	2.153863×10^{-6}	0.999915
27	8.226391×10^{-7}	2.108536×10^{-12}	2.499999×10^{0}	1.331224×10^{-6}	0.999969
28	5.084471×10^{-7}	8.055778×10^{-13}	2.499999×10^{0}	8.227767×10^{-7}	1.000010
29	3.143568×10^{-7}	3.073097×10^{-13}	2.499999×10^{0}	5.084199×10^{-7}	1.000433
30	1.938814×10^{-7}	1.181277×10^{-13}	2.500000×10^{0}	3.145385×10^{-7}	0.997557

Table 8: Super Halley method from $x_0 = 2$ for the double root.

n	$\lvert x_n - x_{n-1}\rvert$	$\lvert f(x_n)\rvert$	x_n	$\lvert x_n - \alpha\rvert$	Rate
0	–	2.900000×10^{-1}	2.000000×10^{0}	5.000000×10^{-1}	–
1	3.706392×10^{-1}	1.981592×10^{-2}	2.370639×10^{0}	1.293608×10^{-1}	–
2	9.689447×10^{-2}	1.252919×10^{-3}	2.467534×10^{0}	3.246630×10^{-2}	1.022485
3	2.434442×10^{-2}	7.847669×10^{-5}	2.491878×10^{0}	8.121881×10^{-3}	1.002343
4	6.091120×10^{-3}	4.907213×10^{-6}	2.497969×10^{0}	2.030761×10^{-3}	1.000368
5	1.523053×10^{-3}	3.067378×10^{-7}	2.499492×10^{0}	5.077078×10^{-4}	1.000078
6	3.807798×10^{-4}	1.917168×10^{-8}	2.499873×10^{0}	1.269280×10^{-4}	1.000019
7	9.519595×10^{-5}	1.198241×10^{-9}	2.499968×10^{0}	3.173209×10^{-5}	1.000004
8	2.379908×10^{-5}	7.488943×10^{-11}	2.499992×10^{0}	7.933012×10^{-6}	1.000003
9	5.949700×10^{-6}	4.680700×10^{-12}	2.499998×10^{0}	1.983313×10^{-6}	0.999977
10	1.487379×10^{-6}	2.922107×10^{-13}	2.500000×10^{0}	4.959336×10^{-7}	0.999868
11	3.709560×10^{-7}	1.776357×10^{-14}	2.500000×10^{0}	1.249776×10^{-7}	0.994391
12	8.704657×10^{-8}	1.776357×10^{-15}	2.500000×10^{0}	3.793102×10^{-8}	0.865094

Table 9: Chord method for $f(x) = 2 - e^x$ with $x_0 = 0$ and $M = 20$.

n	$\lvert x_n - x_{n-1}\rvert$	$\lvert f(x_n)\rvert$	x_n	$\lvert x_n - \alpha\rvert$	Rate
0	–	1.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	–
1	1.000000×10^{0}	7.182818×10^{-1}	1.000000×10^{0}	3.068528×10^{-1}	–
2	7.182818×10^{-1}	6.745949×10^{-1}	2.817182×10^{-1}	4.114290×10^{-1}	-0.359894
3	6.745949×10^{-1}	6.020850×10^{-1}	9.563130×10^{-1}	2.631659×10^{-1}	-1.523697
4	6.020850×10^{-1}	5.749199×10^{-1}	3.542281×10^{-1}	3.389191×10^{-1}	-0.566132
5	5.749199×10^{-1}	5.323505×10^{-1}	9.291479×10^{-1}	2.360007×10^{-1}	-1.430670
6	5.323505×10^{-1}	5.129453×10^{-1}	3.967974×10^{-1}	2.963497×10^{-1}	-0.629148
7	5.129453×10^{-1}	4.836835×10^{-1}	9.097428×10^{-1}	2.165956×10^{-1}	-1.376816
8	4.836835×10^{-1}	4.687886×10^{-1}	4.260592×10^{-1}	2.670880×10^{-1}	-0.668391
9	4.687886×10^{-1}	4.469633×10^{-1}	8.948478×10^{-1}	2.017006×10^{-1}	-1.340009
10	4.469633×10^{-1}	4.350021×10^{-1}	4.478845×10^{-1}	2.452627×10^{-1}	-0.696402
11	4.350021×10^{-1}	4.178690×10^{-1}	8.828866×10^{-1}	1.897394×10^{-1}	-1.312628
12	4.178690×10^{-1}	4.079578×10^{-1}	4.650176×10^{-1}	2.281296×10^{-1}	-0.717873
13	4.079578×10^{-1}	3.940234×10^{-1}	8.729754×10^{-1}	1.798282×10^{-1}	-1.291158
14	3.940234×10^{-1}	3.856184×10^{-1}	4.789520×10^{-1}	2.141952×10^{-1}	-0.735087
15	3.856184×10^{-1}	3.739860×10^{-1}	8.645704×10^{-1}	1.714232×10^{-1}	-1.273703
16	3.739860×10^{-1}	3.667295×10^{-1}	4.905844×10^{-1}	2.025628×10^{-1}	-0.749327
17	3.667295×10^{-1}	3.568217×10^{-1}	8.573140×10^{-1}	1.641668×10^{-1}	-1.259130
18	3.568217×10^{-1}	3.504669×10^{-1}	5.004923×10^{-1}	1.926549×10^{-1}	-0.761383
19	3.504669×10^{-1}	3.418921×10^{-1}	8.509592×10^{-1}	1.578120×10^{-1}	-1.246713
20	3.418921×10^{-1}	3.362616×10^{-1}	5.090671×10^{-1}	1.840801×10^{-1}	-0.771776

Secant

Table 10: Secant method for $f(x) = 2 - e^x$ with $x_0 = 0$, $x_1 = 2$, and $M = 20$.

n	$\lvert x_n - x_{n-1}\rvert$	$\lvert f(x_n)\rvert$	x_n	$\lvert x_n - \alpha\rvert$	Rate
0	–	1.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	–
1	2.000000×10^{0}	5.389056×10^{0}	2.000000×10^{0}	1.306853×10^{0}	–
2	1.686965×10^{0}	6.324302×10^{-1}	3.130353×10^{-1}	3.801119×10^{-1}	-1.947396
3	1.771801×10^{-1}	3.673322×10^{-1}	4.902154×10^{-1}	2.029318×10^{-1}	0.508211
4	2.455090×10^{-1}	8.699318×10^{-2}	7.357244×10^{-1}	4.257718×10^{-2}	2.488149
5	4.700950×10^{-2}	8.845021×10^{-3}	6.887149×10^{-1}	4.432319×10^{-3}	1.448813
6	4.338562×10^{-3}	1.875042×10^{-4}	6.930534×10^{-1}	9.375649×10^{-5}	1.704378
7	9.396442×10^{-5}	4.158722×10^{-7}	6.931474×10^{-1}	2.079361×10^{-7}	1.584871
8	2.079458×10^{-7}	1.949574×10^{-11}	6.931472×10^{-1}	9.747869×10^{-12}	1.631086
9	9.747869×10^{-12}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
10	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–

Status: `error: secant denominator underflow`

Newton

Status: `fixed-iteration run complete`

Super Halley

Status: `fixed-iteration run complete`

3 Experiment 3

Newton from $x_0 = 1$

Status: `fixed-iteration run complete; warning: |f'(x_n)| < TOL`

Newton from $x_0 = -0.705$

Status: `error: derivative underflow in Newton step; warning: |f'(x_n)| < TOL`

Hybrid on $[-0.705, 1]$

Status: `converged`

Table 11: Newton's method for $f(x) = 2 - e^x$ with $x_0 = 0$ and $M = 20$.

n	$\lvert x_n - x_{n-1} \rvert$	$\lvert f(x_n) \rvert$	x_n	$\lvert x_n - \alpha \rvert$	Rate
0	–	1.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	–
1	1.000000×10^{0}	7.182818×10^{-1}	1.000000×10^{0}	3.068528×10^{-1}	–
2	2.642411×10^{-1}	8.706523×10^{-2}	7.357589×10^{-1}	4.261170×10^{-2}	2.422754
3	4.171658×10^{-2}	1.791040×10^{-3}	6.940423×10^{-1}	8.951194×10^{-4}	1.956666
4	8.947189×10^{-4}	8.009998×10^{-7}	6.931476×10^{-1}	4.004998×10^{-7}	1.996413
5	4.004997×10^{-7}	1.603162×10^{-13}	6.931472×10^{-1}	8.015810×10^{-14}	2.000029
6	8.015810×10^{-14}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
7	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
8	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
9	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
10	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
11	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
12	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
13	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
14	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
15	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
16	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
17	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
18	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
19	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–
20	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	0.000000×10^{0}	–

Hybrid on $[-0.71, 2.71]$

Status: `converged`

4 Graduate Section

Naive Bisection Scan

Safeguarded Bisection

Status: `stopped:  sign of p(x_n) is unreliable from roundoff`

5 Discussion

Experiment 1

For the negative root, every method converged rapidly, but the bracketing methods also guaranteed that the root stayed trapped inside the interval. For the double root at $\alpha = 2.5$, the open methods slowed down substantially. This is due to multiplicity, and is observed in the rate column.

Table 12: Super Halley method for $f(x) = 2 - e^x$ with $x_0 = 0$ and $M = 20$.

n	$\lvert x_n - x_{n-1}\rvert$	$\lvert f(x_n)\rvert$	x_n	$\lvert x_n - \alpha\rvert$	Rate
0	–	1.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	–
1	7.500000×10^{-1}	1.170000×10^{-1}	7.500000×10^{-1}	5.685282×10^{-2}	–
2	5.688345×10^{-2}	6.126301×10^{-5}	6.931165×10^{-1}	3.063197×10^{-5}	3.009536
3	3.063197×10^{-5}	9.325873×10^{-15}	6.931472×10^{-1}	4.773959×10^{-15}	–
4	4.662937×10^{-15}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
5	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
6	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
7	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
8	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
9	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
10	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
11	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
12	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
13	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
14	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
15	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
16	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
17	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
18	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
19	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–
20	0.000000×10^{0}	0.000000×10^{0}	6.931472×10^{-1}	1.110223×10^{-16}	–

Experiment 2

The error plot separates the methods by effective order. Newton and Super Halley collapse to machine precision in very few steps, the secant method is slightly above linear, and the chord method performs the worst because it freezes the derivative at $x_0 = 0$. In this test, that fixed slope is not a good linear model near the root, so the method gets stuck.

Experiment 3

This test shows the difference between a purely local method and a safeguarded one. Newton's method is undreliable because the start at $x_0 = -0.705$ sits near a point where $f'(x) = 10(1 - 2x^2)e^{-x^2}$ is almost zero, so the first Newton correction explodes, while the start at $x_0 = 1$ drifts to large positive values where $f(x)$ is numerically tiny even though the iterate is far from the true root $\alpha = 0$. The hybrid method avoids that by falling back to the bracketing interval whenever the Newton step is not trustworthy, so both intervals still converge to the intended root.

Table 13: Newton's method for $f(x) = 10xe^{-x^2}$ with $x_0 = 1$, $TOL = 10^{-4}$, and $M = 30$.

n	$\|x_n - x_{n-1}\|$	$\|f(x_n)\|$	x_n	$\|x_n - \alpha\|$	Rate
0	–	3.678794×10^{0}	1.000000×10^{0}	1.000000×10^{0}	–
1	1.000000×10^{0}	3.663128×10^{-1}	2.000000×10^{0}	2.000000×10^{0}	–
2	2.857143×10^{-1}	1.230424×10^{-1}	2.285714×10^{0}	2.285714×10^{0}	0.192645
3	2.419006×10^{-1}	4.246921×10^{-2}	2.527615×10^{0}	2.527615×10^{0}	0.753363
4	2.146107×10^{-1}	1.486945×10^{-2}	2.742226×10^{0}	2.742226×10^{0}	0.810096
5	1.953207×10^{-1}	5.252669×10^{-3}	2.937546×10^{0}	2.937546×10^{0}	0.844296
6	1.806792×10^{-1}	1.866890×10^{-3}	3.118226×10^{0}	3.118226×10^{0}	0.867518
7	1.690401×10^{-1}	6.664975×10^{-4}	3.287266×10^{0}	3.287266×10^{0}	0.884445
8	1.594813×10^{-1}	2.387607×10^{-4}	3.446747×10^{0}	3.446747×10^{0}	0.897385
9	1.514379×10^{-1}	8.576326×10^{-5}	3.598185×10^{0}	3.598185×10^{0}	0.907628
10	1.445410×10^{-1}	3.087395×10^{-5}	3.742726×10^{0}	3.742726×10^{0}	0.915951
11	1.385374×10^{-1}	1.113453×10^{-5}	3.881263×10^{0}	3.881263×10^{0}	0.922859
12	1.332467×10^{-1}	4.021764×10^{-6}	4.014510×10^{0}	4.014510×10^{0}	0.928689
13	1.285360×10^{-1}	1.454551×10^{-6}	4.143046×10^{0}	4.143046×10^{0}	0.933679
14	1.243051×10^{-1}	5.266623×10^{-7}	4.267351×10^{0}	4.267351×10^{0}	0.938001
15	1.204766×10^{-1}	1.908817×10^{-7}	4.387828×10^{0}	4.387828×10^{0}	0.941783
16	1.169898×10^{-1}	6.924271×10^{-8}	4.504817×10^{0}	4.504817×10^{0}	0.945121
17	1.137961×10^{-1}	2.513734×10^{-8}	4.618614×10^{0}	4.618614×10^{0}	0.948090
18	1.108560×10^{-1}	9.131975×10^{-9}	4.729470×10^{0}	4.729470×10^{0}	0.950748
19	1.081373×10^{-1}	3.319555×10^{-9}	4.837607×10^{0}	4.837607×10^{0}	0.953143
20	1.056133×10^{-1}	1.207367×10^{-9}	4.943220×10^{0}	4.943220×10^{0}	0.955312
21	1.032616×10^{-1}	4.393599×10^{-10}	5.046482×10^{0}	5.046482×10^{0}	0.957286
22	1.010631×10^{-1}	1.599573×10^{-10}	5.147545×10^{0}	5.147545×10^{0}	0.959090
23	9.900184×10^{-2}	5.826043×10^{-11}	5.246547×10^{0}	5.246547×10^{0}	0.960746
24	9.706389×10^{-2}	2.122825×10^{-11}	5.343611×10^{0}	5.343611×10^{0}	0.962271
25	9.523735×10^{-2}	7.737709×10^{-12}	5.438848×10^{0}	5.438848×10^{0}	0.963681
26	9.351183×10^{-2}	2.821349×10^{-12}	5.532360×10^{0}	5.532360×10^{0}	0.964988
27	9.187828×10^{-2}	1.029051×10^{-12}	5.624238×10^{0}	5.624238×10^{0}	0.966203
28	9.032873×10^{-2}	3.754425×10^{-13}	5.714567×10^{0}	5.714567×10^{0}	0.967335
29	8.885617×10^{-2}	1.370150×10^{-13}	5.803423×10^{0}	5.803423×10^{0}	0.968393
30	8.745437×10^{-2}	5.001534×10^{-14}	5.890877×10^{0}	5.890877×10^{0}	0.969384

Table 14: Newton's method for $f(x) = 10xe^{-x^2}$ with $x_0 = -0.705$, $TOL = 10^{-4}$, and $M = 30$.

n	$\|x_n - x_{n-1}\|$	$\|f(x_n)\|$	x_n	$\|x_n - \alpha\|$	Rate
0	–	4.288781×10^{0}	-7.050000×10^{-1}	7.050000×10^{-1}	–
1	1.184874×10^{2}	0.000000×10^{0}	1.177824×10^{2}	1.177824×10^{2}	–

Table 15: Hybrid method for $f(x) = 10xe^{-x^2}$ on $[-0.705, 1]$.

n	a_n	b_n	x_n	$\frac{1}{2}(b_n - a_n)$	$\lvert x_n - \alpha\rvert$
0	-7.050000×10^{-1}	1.000000×10^{0}	-7.050000×10^{-1}	8.525000×10^{-1}	7.050000×10^{-1}
1	-7.050000×10^{-1}	1.000000×10^{0}	1.000000×10^{0}	8.525000×10^{-1}	1.000000×10^{0}
2	-7.050000×10^{-1}	1.475000×10^{-1}	1.475000×10^{-1}	4.262500×10^{-1}	1.475000×10^{-1}
3	-6.710065×10^{-3}	1.475000×10^{-1}	-6.710065×10^{-3}	7.710503×10^{-2}	6.710065×10^{-3}
4	-6.710065×10^{-3}	6.042955×10^{-7}	6.042955×10^{-7}	3.355335×10^{-3}	6.042955×10^{-7}
5	$-4.413042 \times 10^{-19}$	6.042955×10^{-7}	$-4.413042 \times 10^{-19}$	3.021478×10^{-7}	4.413042×10^{-19}

Table 16: Hybrid method for $f(x) = 10xe^{-x^2}$ on $[-0.71, 2.71]$.

n	a_n	b_n	x_n	$\frac{1}{2}(b_n - a_n)$	$\lvert x_n - \alpha\rvert$
0	-7.100000×10^{-1}	2.710000×10^{0}	-7.100000×10^{-1}	1.710000×10^{0}	7.100000×10^{-1}
1	-7.100000×10^{-1}	2.710000×10^{0}	2.710000×10^{0}	1.710000×10^{0}	2.710000×10^{0}
2	-7.100000×10^{-1}	1.000000×10^{0}	1.000000×10^{0}	8.550000×10^{-1}	1.000000×10^{0}
3	-7.100000×10^{-1}	1.450000×10^{-1}	1.450000×10^{-1}	4.275000×10^{-1}	1.450000×10^{-1}
4	-6.364894×10^{-3}	1.450000×10^{-1}	-6.364894×10^{-3}	7.568245×10^{-2}	6.364894×10^{-3}
5	-6.364894×10^{-3}	5.157493×10^{-7}	5.157493×10^{-7}	3.182705×10^{-3}	5.157493×10^{-7}
6	$-2.743328 \times 10^{-19}$	5.157493×10^{-7}	$-2.743328 \times 10^{-19}$	2.578747×10^{-7}	2.743328×10^{-19}

Table 17: Naive bisection on the degree-9 polynomial for slightly perturbed intervals near $[1.92, 2.08]$.

a	b	x_N	N	$\lvert p(x_N)\rvert$
1.920000×10^{0}	2.080000×10^{0}	2.000000×10^{0}	0	0.000000×10^{0}
1.920002×10^{0}	2.079999×10^{0}	2.036987×10^{0}	1000	1.477929×10^{-12}
1.920004×10^{0}	2.079998×10^{0}	2.042266×10^{0}	21	0.000000×10^{0}
1.920006×10^{0}	2.079997×10^{0}	2.005095×10^{0}	36	0.000000×10^{0}
1.920008×10^{0}	2.079996×10^{0}	2.049063×10^{0}	1000	2.785328×10^{-12}
1.920010×10^{0}	2.079995×10^{0}	1.955606×10^{0}	1000	2.160050×10^{-12}
1.920012×10^{0}	2.079994×10^{0}	1.980005×10^{0}	2	0.000000×10^{0}
1.920014×10^{0}	2.079993×10^{0}	1.957796×10^{0}	45	0.000000×10^{0}
1.920016×10^{0}	2.079992×10^{0}	1.982398×10^{0}	1000	2.501110×10^{-12}
1.920018×10^{0}	2.079991×10^{0}	1.981570×10^{0}	8	0.000000×10^{0}
1.920020×10^{0}	2.079990×10^{0}	2.052694×10^{0}	1000	4.774847×10^{-12}
1.920022×10^{0}	2.079989×10^{0}	1.976573×10^{0}	8	0.000000×10^{0}
1.920024×10^{0}	2.079988×10^{0}	2.012517×10^{0}	1000	1.818989×10^{-12}
1.920026×10^{0}	2.079987×10^{0}	1.955169×10^{0}	1000	3.865352×10^{-12}
1.920028×10^{0}	2.079986×10^{0}	2.034935×10^{0}	46	0.000000×10^{0}
1.920030×10^{0}	2.079985×10^{0}	2.016222×10^{0}	1000	5.684342×10^{-13}
1.920032×10^{0}	2.079984×10^{0}	1.950781×10^{0}	42	0.000000×10^{0}

Table 18: Safeguarded bisection with Horner error bounds on $[1.0, 2.5]$.

n	a_n	b_n	x_n	$\frac{1}{2}(b_n - a_n)$	$\lvert\widetilde{p}(x_n)\rvert$	d_n	sign
0	1.000000×10^{0}	2.500000×10^{0}	1.750000×10^{0}	7.500000×10^{-1}	3.814697×10^{-6}	5.861303×10^{-10}	negative
1	1.750000×10^{0}	2.500000×10^{0}	2.125000×10^{0}	3.750000×10^{-1}	7.450581×10^{-9}	1.382064×10^{-9}	positive
2	1.750000×10^{0}	2.125000×10^{0}	1.937500×10^{0}	1.875000×10^{-1}	1.455192×10^{-11}	9.092804×10^{-10}	uncertain

Graduate Section

For $p(x) = (x-2)^9$, naive bisection is extremely sensitive because the polynomial is very flat near the root and Horner evaluation causes catastrophic cancellation. Small changes in the input interval therefore produce noticeably different computed roots. To demonstrate the modified stopping criterion after several bisections, I used the asymmetric interval $[1.0, 2.5]$ for the safeguarded run. The code stops when the Horner error bouind exceeds the magnitude of the computed value, $|\tilde{p}(x_n)| \leq d_n$, since at that point the true polynomial value could be of either sign and bisection's sign test is unreliable.